Briefings in Bioinformatics, YEAR, pp. 1–3
DOI added during production
Published: Date added during production
Paper

On using clustering statistics for assessing plasmid binning tools accuracy

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We discuss the drawbacks of using homogeneity and completeness, statistics defined for the evaluation of clustering accuracy, for evaluating plasmid binning tools accuracy, motivated by the recently published paper "Circling in on plasmids: benchmarking plasmid detection and reconstruction tools for short-read data from diverse species", by Teixeira et al.

Keywords Plasmids detection, clustering, accuracy measure.

Issue section Letter to the Editor

Key Messages

- Plasmid binning bears strong similarity to clustering problems.
- Standard clustering statistics require to be adapted to assess plasmid binning accuracy.
- Accounting for repeats in measuring plasmid binning accuracy is challenging.

For Peer Review

The recent paper "Circling in on plasmids: benchmarking plasmid detection and reconstruction tools for short-read data from diverse species" (Teixeira et al., 2025) benchmarks tools for plasmid binning, a problem motivated by applications in epidemiological surveillance among others. Plasmid binning tools group contigs from draft assemblies into *plasmid bins* with the aim that each bin contains the set of contigs present in a plasmid of the assembled genome. To assess the accuracy of plasmid binning tools, the authors use entropy-based clustering accuracy measures, *completeness* and *homogeneity* (Rosenberg and Hirschberg, 2007; Utt et al., 2014). Given a predicted clustering and a ground truth clustering of the same ground set, homogeneity quantifies the *purity*, i.e. the extent to which a predicted cluster contains elements from several true clusters, while completeness quantifies the *fragmentation*, i.e. the extent to which a true cluster is fragmented in the predicted clusters. Both measures are used to compare the bins generated by plasmid binning tools (predicted bins) against ground truth bins (true bins) obtained from long-reads sequencing data. In this letter, we caution that using homogeneity and completeness to evaluate plasmid binning accuracy suffers from drawbacks, that we outline and illustrate below.

Contigs length. Homogeneity and completeness do ignore the fact that contigs represent sequences of various lengths, which gives more weight to short contigs, that are often numerous in draft assemblies.

Non-plasmid contigs. Generally predicted and true bins do not contain the same sets of contigs, with some true plasmid contigs missing in the predicted bins and some predicted plasmid contigs actually being chromosomal contigs. Teixeira et al. (2025) assign, for a given set of bins (predicted or true), all contigs that do not appear in any bin to a (predicted or true) *chromosome bin*. The predicted and true chromosome bins are generally much larger than the sets of plasmid bins and highly similar to each other, which distorts the homogeneity and completeness toward values that overestimates the actual accuracy of the predicted bins.

Repeat contigs. For the purpose of plasmid binning, a *repeat* is a contig that is shared across multiple (predicted or true) bins; it is well known that plasmid contain repeated sequences that result in such repeats, e.g. Insertion Sequences. So plasmid binning is not a pure clustering problem on a ground set defined as the contigs of a draft assembly, and resolving which copy of a repeat in the predicted bins matches with which copy in the true bins is essential to quantify the accuracy. Teixeira et al. (2025) address this issue by taking the cross-product of repeats between predicted and true bins, and consider each resulting pair as a *pseudo-contig* in the clustering. This leads to the undesirable property that, in the presence of repeats, perfect predicted bins (i.e. identical to the true bins) will not exhibit perfect homogeneity and completeness. For example, if a repeat is present in true bins T_1 and T_2 as well as in predicted bins P_1 and P_2, the pseudo-contigs are the pairs $(T_1, P_1), (T_2, P_1), (T_1, P_2)$ and (T_2, P_2), and at most two pseudo-contigs are considered as correct, while at least two pseudo-contigs are considered as erroneous, decreasing both homogeneity and completeness..

Received: Date Month Year. **Revised received:** Date Month Year. **Accepted:** Date Month Year
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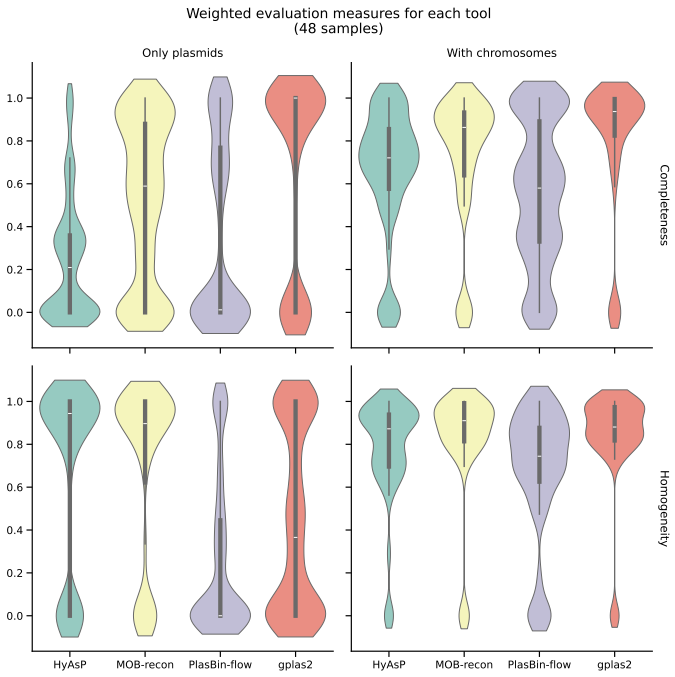

Figure 1 Homogeneity and completeness distributions on 48 sample, fo 4 plasmid binning tools. Top left: completeness with a reduced chromosome bin. Top right: completeness with a full chromosome bin. Bottom left: homogeneity with a reduced chromosome bin. Bottom right: homogeneity with a full chromosome bin.

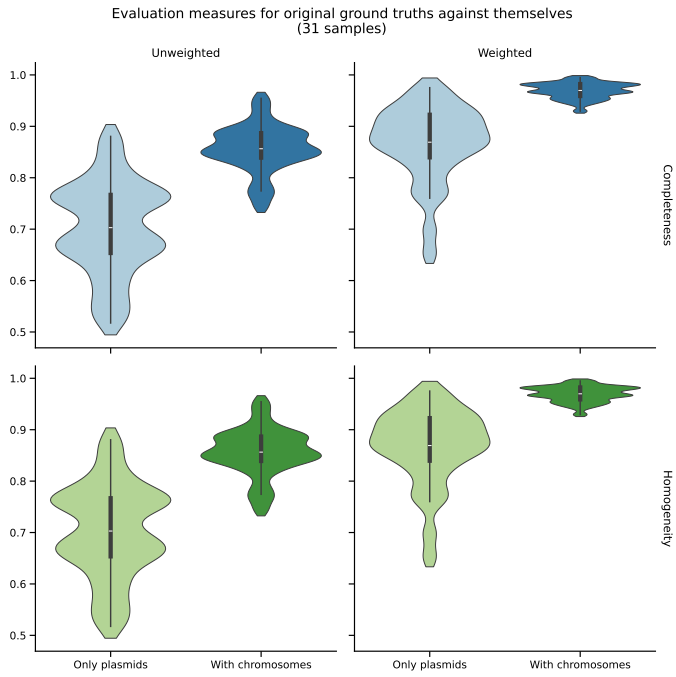

Figure 2 Assessing purity and fragmentation with perfect predicted bins when using the approach of (Teixeira et al., 2025) to handle repeats. Top left: unweighted completeness without (left) and with (right) a chromosome bin. Top right: weighted completeness without (left) and with (right) a chromosome bin. Bottom left: unweighted homogeneity without (left) and with (right) a chromosome bin. Top right: weighted homogeneity without (left) and with (right) a chromosome bin.

Illustration. We illustrate the issues described above with two experiments.

We illustrate the impact of the chromosomal bin on 48 samples considered in (Teixeira et al., 2025). We replaced the chromosome bin as defined in (Teixeira et al., 2025) by a *reduced chromosome bin* excluding all contigs that appear in neither a predicted bin nor a true bin, and computed the corresponding homogeneity and completeness. In this experiment, we account for the length of contigs when computing homogeneity and completeness. Figure 1 shows that using the full chromosome bin biases the homogeneity and completeness toward higher values and can lead to different conclusions as to the respective accuracy of the evaluated tools.

To illustrate the impact of repeats, we considered the set of 31 samples for which the true bins contain repeat(s) and used the true bins as perfect predicted bins, so ensuring by construction perfect purity (resp. fragmentation) of the predicted (resp. true) bins. We provide measures which do (resp. do not) account for the length of contigs, weighted (resp. unweighted) measures. Figure 2 shows that accounting for repeats as in Teixeira et al. (2025) results in homogeneity and completeness that can be far from 1.

Conclusion. While many bioinformatics problems can be seen as clustering problems, they often differ from pure clustering by specific elements that need to be accounted for in adapting or designing accuracy statistics. For assembly-type problems, such as plasmid binning, handling repeats is often challenging both in designing analysis methods and in designing appropriate accuracy measures. To illustrate this, we refer to PlasEval (Mane et al., 2024), a measure of accuracy for plasmid binning that handles repeats by seeking an optimal matching between repeats in predicted and true bins. This can be computationally costly but ensures that perfect predictions result in perfect purity and fragmentation scores. This illustrates the challenge of the balance between the computational footprint required for computing sound accuracy measures and the need to satisfy desirable statistical properties.

Conflicts of interest

The authors declare that they have no competing interests.

Data availability

Code and data to reproduc our experiments are available at `https://github.com/vepain/lte-bib-bbaf589-10-1093`.

Author contributions statement

C.C. designd the study. V.E, A.M. performed implementation and data analysis. V.E., A.M., C.C wrote the manuscript.

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